

FairLens - Bias Audit Report

Fairness, Performance and Explainability Analysis

1. Dataset Summary

Property	Value
Dataset Rows	32537
Dataset Columns	15
Target Column	income
Sensitive Features	race, sex

2. Model Information

Property	Value
Model	Logistic Regression
Task Type	classification
Sensitive Attributes	race, sex

3. Model Performance

Metric	Before Mitigation	After Mitigation
Accuracy	0.8237553779963123	0.822679778733866
Precision	0.812128765589479	0.8110113901036513
Recall	0.8237553779963123	0.822679778733866
F1 Score	0.8086108858683988	0.8062106826411688

4. Fairness Evaluation

Stage	Sensitive Attribute	Metric	Value
original	race	Demographic Parity Difference	0.0000
original	race	Disparate Impact	1.0000
original	race	Equal Opportunity Difference	0.0000
original	race	Group Accuracy	White: 0.8135 Black: 0.8972 Asian-Pac-Islander: 0.8342 Amer-Indian-Eskimo: 0.9434 Other: 0.8837
original	race	Equalized Odds	0.0000
original	sex	Demographic Parity Difference	0.0000
original	sex	Disparate Impact	1.0000
original	sex	Equal Opportunity Difference	0.0000
original	sex	Group Accuracy	Male: 0.7825 Female: 0.9061

Stage	Sensitive Attribute	Metric	Value
original	sex	Equalized Odds	0.0000
mitigated	race	Demographic Parity Difference	0.0000
mitigated	race	Disparate Impact	1.0000
mitigated	race	Equal Opportunity Difference	0.0000
mitigated	race	Group Accuracy	White: 0.8137 Black: 0.8892 Asian-Pac-Islander: 0.8342 Amer-Indian-Eskimo: 0.8868 Other: 0.8837
mitigated	race	Equalized Odds	0.0000
mitigated	sex	Demographic Parity Difference	0.0000
mitigated	sex	Disparate Impact	1.0000
mitigated	sex	Equal Opportunity Difference	0.0000
mitigated	sex	Group Accuracy	Male: 0.7813 Female: 0.9052
mitigated	sex	Equalized Odds	0.0000

5. Bias Assessment

Overall Assessment: No significant bias detected

6. Mitigation Analysis

Recommended Mitigation

Sensitive Attribute	Bias Detected	Recommended Strategy	Reason
race	No	No mitigation required	No significant bias was detected.
sex	No	No mitigation required	No significant bias was detected.

Applied Mitigation

Reweighting

7. Before vs After Comparison

Performance

Metric	Before Mitigation	After Mitigation
Accuracy	0.8237553779963123	0.822679778733866
Precision	0.812128765589479	0.8110113901036513
Recall	0.8237553779963123	0.822679778733866
F1 Score	0.8086108858683988	0.8062106826411688

Fairness

Stage	Sensitive Attribute	Metric	Value
original	race	Demographic Parity Difference	0.0000
original	race	Disparate Impact	1.0000

Stage	Sensitive Attribute	Metric	Value
original	race	Equal Opportunity Difference	0.0000
original	race	Group Accuracy	White: 0.8135
original	race	Group Accuracy	Black: 0.8972
original	race	Group Accuracy	Asian-Pac-Islander: 0.8342
original	race	Group Accuracy	Amer-Indian-Eskimo: 0.9434
original	race	Group Accuracy	Other: 0.8837
original	race	Equalized Odds	0.0000
original	sex	Demographic Parity Difference	0.0000
original	sex	Disparate Impact	1.0000
original	sex	Equal Opportunity Difference	0.0000
original	sex	Group Accuracy	Male: 0.7825
original	sex	Group Accuracy	Female: 0.9061
original	sex	Equalized Odds	0.0000
mitigated	race	Demographic Parity Difference	0.0000
mitigated	race	Disparate Impact	1.0000
mitigated	race	Equal Opportunity Difference	0.0000
mitigated	race	Group Accuracy	White: 0.8137
mitigated	race	Group Accuracy	Black: 0.8892
mitigated	race	Group Accuracy	Asian-Pac-Islander: 0.8342
mitigated	race	Group Accuracy	Amer-Indian-Eskimo: 0.8868
mitigated	race	Group Accuracy	Other: 0.8837
mitigated	race	Equalized Odds	0.0000
mitigated	sex	Demographic Parity Difference	0.0000
mitigated	sex	Disparate Impact	1.0000
mitigated	sex	Equal Opportunity Difference	0.0000
mitigated	sex	Group Accuracy	Male: 0.7813
mitigated	sex	Group Accuracy	Female: 0.9052
mitigated	sex	Equalized Odds	0.0000

8. Explainability

Feature	SHAP Importance
education.num	0.6404
capital.gain	0.5947
age	0.3796
sex	0.3699
marital.status	0.3124
hours.per.week	0.2365
relationship	0.1852
capital.loss	0.1163

Feature	SHAP Importance
race	0.0629
education	0.0565

SHAP importance indicates the relative contribution of each feature to the model's predictions. Higher values indicate greater influence on model behavior.

9. Recommendations

Sensitive Attribute	Bias Detected	Recommended Strategy	Reason
race	No	None	No significant bias was detected.
sex	No	None	No significant bias was detected.

10. Final Conclusion

A mitigation strategy was applied. Review the before and after fairness metrics together with model performance to determine whether the mitigated model provides an appropriate fairness and performance trade-off.

Generated by FairLens - Bias Auditing Toolkit